

Atlanta, GA
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Hunter Ellis

Autonomous Systems Engineer

ellishw.tech
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Skills

Software: C, C++, Python, MATLAB, Simulink, Git, ROS2, Gazebo, Make, CMake, Labview, Qt, PyTorch, OpenCV, ~~Blender~~ ^{Blender}, Verilog, FreeRTOS, Autodesk Inventor (Certified), SolidWorks, Rhino

Hardware: PCB Design and Assembly, Breadboarding, Computer Architecture, Oscilloscope, Multimeter, 3D-Printing

Technical Experience

Avionics Research Engineer

Aug 2025 – Present

Georgia Tech Research Institute · Applied Systems Lab

Atlanta, GA

- Software development for collaborative autonomous UAV swarms.
- Motion planning research for swarm path planning, integrating The Open Motion Planning Library (OPML).
- Collaborative swarm autonomy
- Software systems integration
- embedded software development
- C++ development
- DDS

Robotics Researcher / M.S. Computer Engineering

Aug 2023 – May 2025

Virginia Tech · Control, Optimization, and Online Learning for Autonomy Lab

Blacksburg, VA

- Built a 6-DOF robotic arm and an accompanying ROS2–Gazebo simulation environment for training, evaluating, and deploying custom control algorithms.
- Integrated object detection (YOLOv8), natural language processing, symbolic reasoning, and a DDPG-based reinforcement learning policy for robotic manipulation tasks.
- Aided professors in teaching fundamental concepts in linear systems theory and digital signal processing, including Laplace Transforms, Z-Transforms, system stability, and FIR & IIR filter design.

Thrust Vector Control Intern

May 2024 – Aug 2024

Jacobs Space Exploration Group · Thrust Vector Control Test Lab

Huntsville, AL

- Developed thrust vector control testing hardware and software for NASA's Active Inertial Load Simulator at the Marshall Space Flight Center.
- Created and ran tests to develop a mathematical model of an electro-mechanical actuator – used Python, MATLAB, and LabVIEW.

Control Research Intern

Jun 2023 – Aug 2023

G2ELab · Power Electronics Lab

Grenoble, France

- Researched different inverter control methods for 4-leg (microgrid) topologies in Simulink and HIL simulation.

Naval Research Enterprise Internship Program

Jun 2022 – Aug 2022

NSWC Carderock · Center for Innovation in Ship Design

Bethesda, MD

- Developed concept hospital sea-train designs and estimated fuel consumption and electrical power loads of the concept sea-trains.

Projects

SLAM FPV Drone Simulator

- Drone simulator for training FPV pilots, written in C++ with the Godot game engine.
- Uses SLAM algorithms to create custom environments for pilots to train in.

6-DOF Robot Arm

- 3D printed robot arm, built using stepper motors and pulleys.
- Implemented ROS2 Jazzy control and Gazebo Harmonic simulation.

Education

Master of Science in Computer Engineering

May 2025

Virginia Tech – Software and Machine Intelligence

Blacksburg, VA

Bachelor of Science in Electrical & Computer Engineering

May 2024

Virginia Tech – Control Systems and Machine Learning (double major)

Blacksburg, VA

Papers and Research

A Neuro-Symbolic Reinforcement Learning Architecture: Integrating Perception, Reasoning, and Control – Virginia Tech

Master's Thesis, Jun 2025